

DANIEL GUO

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EDUCATION

University of California, Los Angeles (UCLA), Los Angeles, CA

Expected June 2026

Computer Science

- Relevant course work: Data Structures and Algorithms, Computer Organization, Software Construction, Operating Systems, Computer Graphics
- Awards: Dean's Honor List 2024, Upsilon Pi Epsilon (UPE) Membership

EXPERIENCE

Purrlicious Boba | Full-stack Developer

June 2024 - Present

- Partnered with 5 programmers to design responsive website for local cat cafe and boba shop using TypeScript, Tailwind CSS, and AWS backend services
- Created features for users to find shop information, view drink menu, join newsletter, and contact business

Glocol Networks | Intern

June 2022 - July 2022

- Engineered occupancy detection for mass transit using computer vision and artificial intelligence with Raspberry Pi modules
- Extracted Amtrak train departure and arrival times using JavaScript and displayed information on frontend using Vue.js, HTML, and CSS

PROJECTS

BruinBowl | Full-stack Developer

March 2024 - June 2024

- Collaborated with 6 students to design interactive web application, allowing users to quiz themselves on UCLA trivia, while managing version control with Git
- Designed user and gameplay functionality using MERN stack; created secure user login and authentication in backend alongside features scalable for hundreds of users: question search, real-time leaderboards, and user profiles

UCLA ACM AI | Projects Member - Emoji GAN

January 2024 - June 2024

- Worked with 3 team members to apply Python, PyTorch, and NumPy skills to design a dataset, data loader, and convolutional neural network for organizing and processing emoji image data
- Implemented generator, discriminator, and training loop to develop a novel emoji GAN model for expressive image generation and discrimination, able to generate and display increasingly realistic emojis

UCLA ACM Studio | Project Rudd - Programmer

January 2024 - June 2024

- Collaborated with 7 team members to build and test a videogame in style of 2D platformer using Unity 2D and C#
- Drafted and deployed movement, collision detection, momentum, gravity, and integrated cutscenes while maintaining version control through Git, reviewing over 50 commits for bugs

RESEARCH EXPERIENCE

Trustworthy AI Lab | Student Researcher

March 2024 - Present

- Analyzed and enhanced data loading efficiency of CTGAN (tabular data synthesizer) by 100 times through partitioning of training data with Dask Delayed objects while ensuring compatibility with PyTorch tensor datasets
- Utilized PyTest to rigorously test column transformers and reverse transformations, ensuring model integrity and reliability throughout pipeline, while packaging Python library using Docker to avoid environment inconsistencies

Cyber Summer Research Fellowship | Cybersecurity Researcher

July 2024 - September 2024

- Developed convolutional neural network using PyTorch to detect image steganography (LSB, PVD, DCT, FFT), achieving over 87% accuracy in multiclass classification
- Drafted and implemented methods for argument parsing, model freezing and unrolling, and displaying images from dataset of over 33,000 clean and steganographically altered images
- Co-authored Deep Learning for Steganographic Image Detection, accepted by 2024 IEEE MIT Undergraduate Research Technology Conference (URTC)

COMPUTER SKILLS

- Programming Languages: Python, C++/C#/C, JavaScript, HTML, CSS, Java, SQL, Bash, Verilog
- Frameworks and Libraries: Vue.js, MongoDB, Express.js, React, Node.js, PyTorch, NumPy, Pandas, Dask
- Other: Git, Docker, Unity, SolidWorks

ACCOMPLISHMENTS

- Awards: Class of 2023 Valedictorian (Vista del Lago High School), CSF Lifetime Membership, NHS Lifetime Membership, AP Scholar with Distinction (2022-23), AP Scholar with Honor (2021)